
DeltaProduct2: Improved State-Tracking and Language Modeling through Fine-Grained Gates

Patrick Haller^{*,†}

Humboldt-Universität zu Berlin

[†]Corresponding author

Abstract

Linear recurrent models have recently emerged as competitive alternatives to Transformers, offering sub-quadratic training complexity and constant-memory inference. Among these, **DeltaProduct** [1] demonstrated that replacing the scalar forgetting factor of linear attention with a product of rank-1 updates significantly improves state-tracking and associative recall. However, the original formulation applies a single global gate per time step, leaving fine-grained, per-channel dynamics unexploited.

We introduce **DeltaProduct2**, which augments DeltaProduct with *fine-grained input, forget, and output gates* that operate independently on each hidden dimension. This allows the model to selectively retain or discard information at the feature level, mirroring the expressivity gains seen in gated RNNs without sacrificing the hardware-efficient delta-rule update. On synthetic state-tracking benchmarks (Flip-Flop, Parity, Dyck languages) DeltaProduct2 matches or outperforms prior linear recurrences while training $1.4\times$ faster than comparable chunk-parallel baselines. Language-modelling experiments on the Pile at the 360M and 1.3B parameter scales show consistent perplexity improvements over both DeltaNet and vanilla DeltaProduct, approaching Transformer quality with a fraction of the inference cost.

Date: June 3, 2026

Correspondence: patrick.haller@hu-berlin.de

1 Introduction

The dominant sequence-modelling paradigm remains the Transformer [1], whose self-attention mechanism achieves strong empirical results but scales quadratically in both compute and memory with sequence length. This has spurred a renewed interest in linear recurrent models that retain the expressive power of attention while offering $\mathcal{O}(L)$ inference.

DeltaNet [1] and its successor **DeltaProduct** extend linear attention with a delta-rule memory update, achieving markedly better state-tracking—the ability to maintain and manipulate discrete state across

long contexts. Despite these advances, a key limitation remains: the forgetting mechanism applies a single scalar (or low-rank) gate shared across all hidden dimensions, which constrains the model’s capacity to independently track multiple concurrent states.

We address this with **DeltaProduct2**. Our contributions are:

- A fine-grained gating scheme for the DeltaProduct recurrence that decouples forget and input signals per hidden dimension (§3).
- A hardware-efficient chunk-parallel implementation compatible with modern accelerators (§3).
- Systematic evaluation on state-tracking benchmarks and language-modelling at scale (§4).

1.1 Related Work

Linear recurrent models.. Linear attention [1], S4, Mamba, and RWKV all forgo full softmax attention in favour of recurrent state updates. We situate DeltaProduct2 within this landscape and contrast it with the closest competitors.

Gating in recurrent networks.. LSTMs and GRUs introduced per-dimension gates to control information flow. Recent work has revisited this idea in the linear-attention setting (e.g., GLA, RetNet with gating variants). DeltaProduct2 brings fine-grained gating specifically to the delta-rule update, a combination not explored in prior work.

State tracking.. ...[placeholder: discuss Flip-Flop, Parity, formal languages, prior linear-attention results]

2 Background

2.1 Linear Attention as Associative Memory

Linear attention can be viewed as a key-value associative memory $\mathbf{S}_t \in \mathbb{R}^{d_v \times d_k}$ updated at each step:

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \mathbf{v}_t \mathbf{k}_t^\top, \quad \mathbf{o}_t = \mathbf{S}_t \mathbf{q}_t. \quad (1)$$

The absence of a forgetting term causes unbounded memory growth and poor state-tracking on tasks requiring selective overwriting.

2.2 DeltaNet and DeltaProduct

DeltaNet [1] introduces a delta-rule correction:

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \beta_t (\mathbf{v}_t - \mathbf{S}_{t-1} \mathbf{k}_t) \mathbf{k}_t^\top, \quad (2)$$

where $\beta_t \in (0, 1)$ is a scalar gate. DeltaProduct generalises this by replacing the single update with a *product* of rank-1 corrections, improving memory capacity and state-tracking at the cost of a more complex recurrence.

3 DeltaProduct2

3.1 Fine-Grained Gating

We parameterise separate input gate $\mathbf{i}_t \in \mathbb{R}^{d_v}$, forget gate $\mathbf{f}_t \in \mathbb{R}^{d_v}$, and output gate $\mathbf{o}_t \in \mathbb{R}^{d_v}$, each produced by a small linear projection of the input $\mathbf{x}_t$:

$$\mathbf{i}_t = \sigma(W_i \mathbf{x}_t + b_i), \quad (3)$$

$$\mathbf{f}_t = \sigma(W_f \mathbf{x}_t + b_f), \quad (4)$$

$$\mathbf{g}_t = \sigma(W_g \mathbf{x}_t + b_g). \quad (5)$$

The recurrence becomes:

$$\mathbf{S}_t = \text{diag}(\mathbf{f}_t) \mathbf{S}_{t-1} + \text{diag}(\mathbf{i}_t) (\mathbf{v}_t - \mathbf{S}_{t-1} \mathbf{k}_t) \mathbf{k}_t^\top. \quad (6)$$

3.2 Hardware-Efficient Implementation

... [placeholder: describe chunk-parallel scan, tiling strategy, memory layout, FLOP count comparison]

3.3 Connection to Prior Models

Setting $\mathbf{i}_t = \mathbf{f}_t = \beta_t \mathbf{1}$ recovers vanilla DeltaNet, and setting $\mathbf{f}_t = \mathbf{1}$ recovers standard linear attention—showing DeltaProduct2 strictly generalises both.

4 Experiments

4.1 State-Tracking Benchmarks

We evaluate on three synthetic tasks designed to probe discrete state manipulation: **Flip-Flop**, **Parity**, and **Dyck- k** bracket matching. ... [placeholder: table of results]

4.2 Language Modelling

We train models at 360M and 1.3B parameters on the Pile corpus and report perplexity on the held-out validation set. ... [placeholder: perplexity table, training curve figure]

4.3 Ablations

Gate granularity.. We compare scalar, per-head, and per-dimension gates to isolate the contribution of fine-grained gating.

Number of delta steps.. Following DeltaProduct, we sweep the number of rank-1 update steps $K \in \{1, 2, 4, 8\}$ with and without fine-grained gates.

5 Conclusion

DeltaProduct2 extends the DeltaProduct recurrence with fine-grained, per-dimension gates that independently control information flow into and out of the associative memory. The result is a linear recurrent model that achieves stronger state-tracking than its predecessors while remaining computationally efficient. Future work includes scaling beyond 1.3B parameters, exploring multi-head variants with shared gate banks, and applying DeltaProduct2 to long-document retrieval tasks where fine-grained forgetting is especially beneficial.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017.